



Micro and nano cellular amorphous polymers (PMMA, PS) in supercritical CO₂ assisted by nanostructured CO₂-philic block copolymers – One step foaming process

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ABSTRACT

Microcellular foaming of commodity amorphous polymers, poly(methyl methacrylate) (PMMA), and poly(styrene) (PS) was studied in supercritical CO₂ via a batch one-step process in the presence of block copolymers able to change their foaming behaviour and therefore the porous structures. Triblock (styrene-co-butadiene-co-methylmethacrylate SBM, methylmethacrylate-co-butylacrylate-co-methylmethacrylate MAM) terpolymers were blended to PS or PMMA by extrusion. They showed advantages compared to classical PS-PMMA polymer blends in terms of cell size control and reduction of cell size. Foaming is carried out on bulk injection molded samples which were saturated under high pressures of CO₂ (300 bars) at different temperatures (25 °C to 80 °C) and different depressurization rates (pressure drop rates from 150 bar/min to 12 bar/min). Very distinct cellular structures and densities were controlled by varying either the copolymer type or the foaming conditions (T,P). Cell sizes ranged from 0.2 μm to 200 μm, and densities from 0.30 g/cm³ to 1 g/cm³ in the polymers considered. Particularly, when triblock copolymers were able to self organize (nanostructuring) in a polymer matrix, they became phase separated at a nanometer level, presenting nanostructured polymers matrixes. To conclude the study, a possible nanostructuring mechanism is suggested based on the interplay between rubbery and highly CO₂-philic blocks/rigid and less CO₂-philic blocks. It is demonstrated that block copolymer additives are a good pathway towards micro and ultra microcellular supercritical CO₂ foaming of amorphous polymers.

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1. Introduction

Supercritical carbon dioxide (scCO₂) is well known as a polymer foaming agent, used both in industrial processes and in polymer research [1–5]. The foaming processes usually lead to cellular plastics which are most of the time poorly controlled. In the framework of using scCO₂ for producing advanced polymer materials, either one-step or two-step processes can be suggested. Several automotive and industrial applications have been developed recently under the Mucell and Trexel technologies.

In previous works, two-step foaming process amorphous polymers and blends have been deeply investigated [6,7]. In the present work, we will analyze the one-step scCO₂ process on amorphous commodity polymers (PS, PMMA), in view of enlarging the acces-

sible cell-size range towards ultramicro-cellular or nano-cellular polymers using nano dispersed CO₂ philic additives.

There are two main batch routes commonly employed in microcellular foaming processes. In the first process, denominated one-step process, the polymer is saturated with CO₂ in the supercritical regime, during a fixed time. After saturation, the sample, approaching or being in a rubbery state, is depressurized to atmospheric pressure, taking advantage of the swelling and plasticization of the polymer, which reduces the glass transition temperature, allowing the gas expansion. In this method, the cellular microstructure may be controlled by changing the saturation temperature and the depressurization rates. Poly(styrene) [8–10], polycarbonate [11,12], poly(styrene)-co-methylmethacrylate [13,14], poly(methyl methacrylate) [15–17] and fluorinated ethylene-propylene copolymers [18] have been already obtained as microcellular polymers. In the second process, denominated two-step process, the polymer is saturated with ScCO₂ at high pressures and rather low temperatures. Next, the polymer/gas mixture is quenched into a glassy supersaturated state by reducing the pressure to atmospheric value. Finally, in the sec-

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ond step, after removing rapidly the sample from the CO₂ vessel, polymer is foamed by heating to a temperature above or near the glass transition temperature, leading to nucleation and cell growth. Foaming temperature and foaming time are the main parameters to adjust the cellular structure of the final product [19,20] leading to macro or micro cellular polymers. Cellular materials of poly(styrene) [21], poly(propylene) [22,23], poly(ethersulfone) and poly(phenylsulfone) [24,25], poly(carbonate) [26], poly(methyl methacrylate) [27] and poly(lactic acid) have been prepared [28].

In the present work, only the one-step process will be used. The objective is to investigate how the cell size (expressed by an average diameter ϕ , μm), the cell density (N_c expressed by the number of cells/cm³) and the foam density (ρ_f , g/cm³) could be triggered especially towards ultramicro and nano cellular materials, i.e. by decreasing cell size and/or increasing cell density in amorphous polymers, thanks to the addition of well selected nano dispersed copolymers.

Several works have deal with the effect of the addition of polymeric additives in the microcellular foaming behaviour [29,30]. In this sense, polymeric “additives” are considered in a broad sense and are introduced in the following directions:

- Structuration (self assembly) of a neat block copolymer with a highly CO₂-philic and rubbery block and a less CO₂-philic rigid block followed by CO₂ foaming where preferential foaming occurs in the CO₂ philic blocks.
- Blending of such block copolymers in homopolymers, where location of the CO₂-philic blocks is provided at a nanometer level in the homopolymer matrix by nanostructuration of CO₂-philic domains.

2. Experimental

2.1. Materials

2.1.1. Homopolymers

- (1) High impact transparent poly(methyl methacrylate) (PMMA) was kindly supplied by Altuglas Company-Arkema (France) in the form of pellets.
- (2) High molecular weight transparent poly(styrene) (PS Crystal) was kindly supplied by Total Petrochemicals company (France), in the form of pellets.

2.1.2. Copolymers and blends

All block copolymers, were kindly supplied by Arkema Company (France), in the form of pellets, and used as received.

- (1) Styrene–butadiene–methylmethacrylate terpolymer (poly(styrene)-block-polybutadiene-block-poly(methyl methacrylate) named SBM) was selected. Composition of the triblock copolymer was as follows: 52 wt% styrene–30 wt% butadiene–18 wt% methylmethacrylate with a PS block of molar mass equal to 30,000 g/mol.
- (2) Methylmethacrylate–butylacrylate–methylmethacrylate terpolymer (poly(methyl methacrylate)-block-poly(butylacrylate)-block-poly(methyl methacrylate) named MAM) with 35 wt% methacrylate–30 wt% butylacrylate–35 wt% methacrylate, with a PMMA block of molar mass equal to 90,000 g/mol.

Two blends were produced in order to analyze the effect of the additives in the foaming behaviour. First, a PS+SBM blend, with 90 wt% poly(styrene) and 10 wt% SBM, was obtained. Secondly, a PMMA+MAM blend, with 90 wt% PMMA and 10 wt% MAM was

obtained. In both cases, blending by extrusion was carried out, and final materials were obtained in the form of pellets.

Table 1 shows the physical properties of the materials employed. Average molar masses (M_w) were determined by size exclusion chromatography in a PL-GPC50Plus device using THF (Tetrahydrofuran) as a solvent with a concentration of 3 mg/ml with UV detection at 254 nm. Calibration of the GPC equipment was carried out using trichlorobenzene as flow marker, with a constant flow of 1.0 ml/min, using THF as a solvent and a standard poly(styrene) as a calibration material. Glass transition temperature (T_{g0}) was measured in a DSCQ100 equipment, using a heating cycle from -50°C to 200°C at $5^\circ\text{C}/\text{min}$. The T_g detected, those mentioned in Table 1, does not necessarily correspond to the T_g of neat blocks; e.g. T_g of the soft block could not be measured in our conditions, as already reported by other works [31,32]. The T_g values are mentioned as detected by DSC.

2.2. Sample preparation

Blends (PS+SBM or PMMA+MAM) were produced as follows: materials were dried in vacuum (680 mm Hg), at 80°C during 4 h before processing. Mixing by extrusion was carried out using a Scamex CE02 extruder, with a temperature profile from 165°C to 225°C , at a screw velocity of 60 rpm.

In the second step, pellets were injected into sheets (50 mm $\times$ 15 mm) with 3 mm thickness, using a small scale injection moulding machine developed by DSM Xplore. The working temperature was fixed at 240°C , whereas mould temperature was 60°C . The injection pressure was fixed at 12 bar. Samples obtained showed a good surface appearance as well as a good injection behaviour, with no presence of air bubbles inside the sheet.

Concerning nanostructuration, neat block MAM or SBM copolymers are known to have a robustness of self-assembly. Nanostructures are perfectly defined in solvent cast specimens and, although more poorly, still defined in bulk injected samples [33]. As seen in our first work, in PMMA+MAM blends, structuration of MAM in PMMA is also observed whatever the PMMA content and the polymer dispersity. In this case, the PMMA ratio provided worm-like nanostructures in the bulk PMMA–10%MAM injected blends.

2.3. scCO₂ foaming

All the foaming experiments were performed in a high pressure reactor provided by TOP Industry (France), with a capacity of 300 cm³, operating at different temperatures and a constant pressure of 300 bar. During all the process, the temperature and pressure were controlled automatically by a internal software provided together with the equipment. The reactor was built in stainless steel, with a cylindrical vessel of 6 cm diameter and 10 cm of depth. Wall thickness was 10 cm to avoid the gas scape due to leaks or imperfections in the vessel.

As explained before, saturation pressure was fixed at 300 bar for all the experiments, using different foaming temperatures (from 25°C to 80°C) and different foaming times (from 12 bar/min to 150 bar/min). The temperature values are reported to have a rough idea of the sample temperature during the foaming process although the observed temperature drop is dependent on the vessel volume and on the pressure drop rate, so that the measured temperature does not represent the real sample temperature due to poor thermal conductivity of polymers. However the temperature–time plots can be used to evaluate the vitrification of the samples, and to monitor the temperature during the depressurization process.

Thus, a collection of experiments were performed in a one-step batch process. In a first row of experiments depressurization time was fixed (60 bar/min), varying the saturation temperature

Table 1
Physical properties of the materials employed before foaming.

Material	Glass transition (°C)	Molecular weight (g/mol)	Bulk density (g/cm ³)	Depressed glass transition (°C)
PS	103	140,000	1.05	48
PMMA	111	70,000	1.17	52
PS + SBM	98	30,000	1.01	39
PMMA + MAM	109	80,000	1.02	43
MAM	95	98,000	1.08	40
SBM	96	30,000	1.01	34

(25 °C, 40 °C, 50 °C, 60 °C, 70 °C and 80 °C). In this procedure, samples were saturated with supercritical CO₂ at a maintained pressure of 300 bar during 16 h. The saturation time was selected to assure the equilibrium of CO₂ in the polymer. Foaming was carried out by depressurization at a constant rate of 60 bar/min.

In a second row of experiments, saturation temperature was fixed, and depressurization rates were varied. As in the previous experiments, saturation conditions were 300 bar during 16 h, whereas saturation temperature was fixed at 40 °C. Experiments were carried out at different depressurization rates (150 bar/min, 75 bar/min, 50 bar/min, 20 bar/min and 12 bar/min). During the foaming process, due to the pressure drop, temperature inside the reactor was reduced drastically depending of the initial temperature and the pressure drop rate. In our case, the temperature drop inside the reactor was strongly dependent on the depressurization rate. Samples were then removed from the vessel after reaching room temperature and kept 10 min before the density measurements.

2.4. Foam density

Foam density ρ_f was determined by water-displacement method, based on Archimedes principle. Densities were calculated by measuring the volume of water displaced by the sample divided into the sample mass. At least three measurements were carried out for each sample produced.

2.5. SEM observations

Cellular structure was analysed by means of Scanning Electron Microscopy (model HITACHI S-3000N). For the preparation of the samples, foams were frozen in liquid nitrogen and fractured to assure that the microstructure remained intact. For the observations, surfaces were coated with gold using a sputter coater (model EMScope SC 500), in argon atmosphere. The micrographs obtained were analysed to measure cell sizes and assign the presence of a solid unfoamed outer skin. Average cell size ϕ was obtained from direct observation, using a minimum of 100 cells in each calculation, whereas cell density N_c (cells/cm³) was calculated from the next equation [34]:

$$N_c \cong \left(\frac{nM^2}{A} \right)^{3/2}, \quad (1)$$

where n is the number of cells observed in a SEM micrograph, A is the area of the micrograph in cm² and M is the magnification factor.

3. Results and discussion

In a gas process for polymer foaming, cellular structures usually result from a balance between three different basic mechanisms:

- Diffusion of CO₂ in the polymer.
- Affinity to CO₂ (solubility).
- Viscoelastic and rheological behaviour and the glass-to-rubber transition (T_g). Saturation (and probably nucleation) can occur

either in a glassy state or in a rubbery state, but growth can mainly occur in a rubbery state or near the T_g zone of an amorphous polymer.

Previous works on high-pressure gas–polymer interactions [35] show that T_g is classically depressed in a gas or a supercritical liquid. For example, T_g of PS drops from 95 °C to 80 °C after saturation under 30 MPa at 35 °C in CO₂, whereas T_g of PMMA drops even more, from 110 °C to less than 70 °C in the same conditions. This depression is often linear with pressure applied according to the Chow's law [36].

3.1. Sorption of CO₂/swelling and T_g depression vs. vitrification

The global absorption kinetics of CO₂ at a given conditions (300 bar, 40 °C) were measured experimentally. Experimental data were obtained removing the samples and transferring them after 10 min to a high precision balance to record the increase in weight due to gas absorption. This value was selected to assure the complete exchange between the CO₂ inside the samples and the atmospheric air due to diffusion process. Although a different plasticization effect can be observed in different polymers, we chose the same time for both polymers to compare the results. Fig. 1 presents the resultant saturation curves for all the materials under study.

As it can be seen in Fig. 1, the equilibrium situation in the gas–polymer system is completely reached after 16 h of saturation (300 bar, 40 °C). The plateau value of CO₂ uptake w (see Fig. 1) is quantifying the affinity of CO₂ for the material considered; this is the first thermodynamic parameter to be considered for a gas foaming process. This quantity plays a role on the final foam density by plasticization of the polymer, lowering the T_g of the polymer due to CO₂ swelling, which in turn delays the vitrification of the polymer during foaming, thus allowing more time for cell growth. PMMA and PMMA based copolymer are, as expected, the most CO₂-philic polymers in the list chosen.

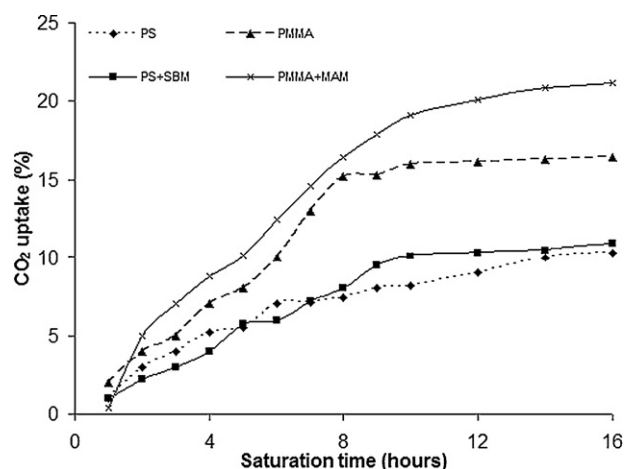


Fig. 1. CO₂ saturation curves for materials analysed (300 bar, 40 °C).

Table 2

Depressed glass transition temperatures according to Hwang's law [36] (saturation conditions 300 bar, 40 °C, 16 h).

Material	Glass transition (°C)	Depressed glass transition (°C)
PS	103	48
PMMA	111	52
PS + SBM	98	39
PMMA + MAM	109	43
MAM	95	40
SBM	96	34

Theoretical models to predict the depression of the glass transition temperature have been proposed. An expression has been developed by Hwang et al. [37]. In his investigation, the relationship between gas absorption and the glass transition temperature in a batch foaming process is analysed, measuring the change in the elasticity modulus induced near the glass transition temperature.

$$T_g = T_{g0} \exp[-(M_p)^{-1/3}(\rho_s)^{-1/4}\alpha \cdot w], \quad (2)$$

where α is a material constant determined by both polymer and gas, M_p is the weight average molar mass of the polymer, w is the weight fraction of CO₂ absorbed during the saturation process, T_{g0} is the glass transition temperature of the polymer and ρ_s is the specific density of the polymer.

Theoretical predictions of this model can be calculated considering different α constants. The rest of the experimental parameters employed in the equation can be determined experimentally. Glass transition temperature T_{g0} was measured by means of DSC technique, the specific gravity of the polymer ρ_s is measured using a high precision balance, the molecular weight of the polymer M_w were measured by SEC (see Table 1), and finally the weight fraction of CO₂ absorbed by the polymer w was determined as explained previously (see Section 3.1). Table 2 presents the theoretical predictions of T_g after saturation at 40 °C and 300 bar using $\alpha = 0.7$ the typical value employed in polymer–scCO₂ mixtures at those conditions [37]. The model is here applied for multiphase materials and only using one T_g (the main one detected by DSC), thus omitting the T_g of soft blocks. However, we believe it provides reasonable data due to a low content of copolymer (10%). Also this model was applied to PMMA at high CO₂ pressures in the supercritical regime [38] in spite of its specific retrograde behaviour [39–44] negligible at these pressures.

In Table 2 the absorption values concerning the scCO₂ process are summarized: the weight fraction of CO₂ absorbed (w) from the experimental results shown in Fig. 1, the depressed glass transition temperature (T_g) derived from the prediction of Hwang's model [36]. It is important to notice that this is only an estimation of the depressed glass transition temperature (T_g), and a direct measurement of this value can only be performed using magnetic suspension balances. Weight of CO₂ absorbed was measured by quickly transferring the samples from the vessel to a high-precision balance.

From the results listed in Table 2, it is clear that all the polymers exhibited a remarkable plasticization effect, with glass transition temperatures being lowered down to the range of 34–52 °C. For example, in the diblock copolymer SM, the T_g descends from 115 °C to 52 °C, with a weight gain of CO₂ of 12%. In the case of the triblock SBM, the T_g is lowered from 96 °C to 34 °C, with a lower weight gain of CO₂ (8.1%). On the other hand, the triblock MAM presents the highest CO₂ affinity ($w = 22.9\%$), with a similar plasticization effect, reducing the T_g from 95 °C to 40 °C. The glass transition temperature of the gas–polymer system is a key parameter to set the polymeric viscoelastic behaviour (glassy, liquid, ...), which determines the final cellular structure of the material. However, the estimated T_g depression seems to be a rather stable parameter for a given polymer system, only dependent of the saturation conditions

Table 3

Weight gain ratio of CO₂ at different saturation temperatures (saturation conditions 300 bar, 16 h).

Material	CO ₂ uptake (%)					
	25 °C	40 °C	50 °C	60 °C	70 °C	80 °C
PS	10.4	10.3	10.1	9.4	9.2	8.9
PMMA	16.9	16.4	16.2	16.1	15.9	15.6
PS + SBM	12.1	10.9	10.5	10.4	10.2	9.7
PMMA + MAM	22.3	21.2	20.4	20.1	19.3	19.2
MAM	24.1	22.9	22.4	22.4	22.1	21.9
SBM	8.3	8.1	8.0	7.6	7.6	7.4

(i.e. 300 bar, 40 °C). An explanation is that in Eq. (2), molar mass and w are counterbalanced.

To obtain complete behaviour of the weight gain of CO₂ into the polymer, saturation experiments were carried out at different temperatures, in a similar way as showed in Fig. 1. Samples were saturated at 300 bar for 16 h at temperatures of 25 °C, 50 °C, 60 °C, 70 °C and 80 °C. The results obtained are presented in Table 3.

As it can be seen from the data in Table 3, weight gain ratio w decreased as saturation temperature increased. For example, at a saturation temperature of 25 °C, predicted T_g of the PS was 46 °C, whereas at 80 °C, T_g is 52 °C. Differences in depressed T_g were no greater than 7 °C. These values indicate that the plasticization effect can be considered nearly similar for all the saturation temperatures employed and the estimated vitrification thresholds of the polymer–gas systems are comparable. However, when the two different foaming routes were analysed, polymer–gas systems show to have different vitrification capabilities.

In the first route, for a fixed saturation temperature, cells are formed by triggering the depressurization rate from 150 bar/min to 12 bar/min, thus inducing a pressure drop ΔP , from the same initial quantity of swollen CO₂ in samples of exactly the same geometry. In the second route, depressurization rate is fixed at 60 bar/min, whereas cells are formed from a saturation temperature between 25 °C and 80 °C.

Fig. 2 presents the temperature versus time variations for both routes. The extent of this temperature drop depends on factors such as reactor volume, thermal conductivities, sample mass or pressure drop rate. Nevertheless, for the “ ΔP route”, the curve exhibits a sudden temperature drop (Fig. 2a).

So, it is proved that temperature inside the reactor may be decreased greatly: down to –50 °C at the end of the process (for a depressurization rate of 150 bar/min), and down to –10 °C for the lowest depressurization rate (12 bar/min). Therefore, the vitrification time induced by the pressure drop is modified and occurs much earlier if ΔP is increased, freezing the cells.

In the second route of experiments, the fixed depressurization rate of 60 bar/min induces a moderate decrease in the temperature drop inside the reactor of 20–40 °C with respect to the initial temperature).

3.2. Cellular morphologies vs. copolymer

3.2.1. Behaviour of homopolymers – PMMA and PS

In Fig. 3, different cell sizes are obtained with an amazing comparable density (between 0.8 and 1 g/cm³) for PMMA. The mechanism involves only the time for the polymer to become glassy and to freeze the cell size.

For a fixed saturation temperature of 40 °C (Fig. 3b), varying depressurization rates did not lead to different cell sizes, which obtained an average cell size of 2 μ m in all the foams produced using this route. However, in Fig. 3a it can be seen that bigger cell sizes are obtained at higher saturation temperatures, because increasing saturation temperature leads to a higher plasticized

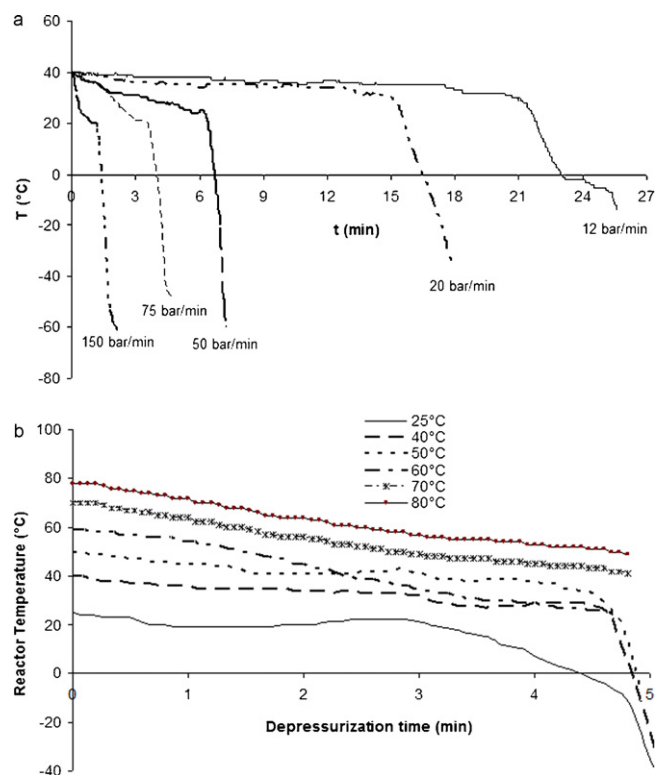


Fig. 2. Typical temperature and pressure drop curves, in the one-step CO_2 processes. (a) Saturation temperature 40 °C, variable depressurization rate. (b) Fixed depressurization rate 60 bar/min, saturation temperature from 30 °C to 80 °C.

state respect to lower temperatures. Moreover, assuming that the Hwang's model approximation to estimate the depressed T_g is accurate, we can state that microcellular foaming is possible even at temperatures below the depressed T_g (e.g. foaming at 25 °C), which

means that the CO_2 is able to expand in a quasi solid state with a low degree of plasticization but at a high CO_2 concentration.

In addition, an outer dense skin may be observed depending on the initial saturation temperature T_{sat} , dense skins at $T_{\text{sat}} < 40$ °C while the skin is suppressed above 50 °C. Such closed outer skin was described in our previous work [6]. This is again correlated to the vitrification phenomenon, the higher the initial temperature is, the more time CO_2 has to diffuse out of the sample while generating cells. At lower T_{sat} the material enters a glassy zone earlier and consequently the outer parts of the sample (that are the first zones to release gas) cannot undergo the nucleation-growth (NG) stages. Gas escapes by diffusion without NG. In other words, during the ΔP step, the NG mechanism is hindered selectively from the outer to the inner layers. Therefore the growth stage is hindered or even suppressed; expansion is here controlled mainly by time of diffusion of CO_2 before vitrification.

The change to PS homopolymer is characterized by a decrease in the affinity for CO_2 (Fig. 2a) and thus a much greater difficulty for foaming in a one-step CO_2 process. Indeed PS does not foam in the conditions used for PMMA (300 bar, 40 °C, $\Delta P/\Delta t = 150$ bar/min) but foaming begins only at higher saturation temperatures (above 60 °C) and lower depressurization rates (below 50 bar/min). It is important to remark that samples processed at 150 bar/min at 40 °C and 60 bar/min at room temperature were not foamed, and SEM micrographs will not be presented here. Fig. 4 presents some SEM micrographs obtained showing the morphology evolution for the rest of the microcellular PS foams produced.

In Table 4 the main structural parameters are presented. Average cell size (ϕ), cell density (N_c) and density (ρ_f) for the two homopolymers.

3.2.2. Behaviour of blends/role of CO_2 philic nano dispersed copolymers

Structuration is retained when blending with homopolymers. In our previous work, several AFM images (operating in tapping mode) of PMMA and PS based blends were presented [6]. Consid-

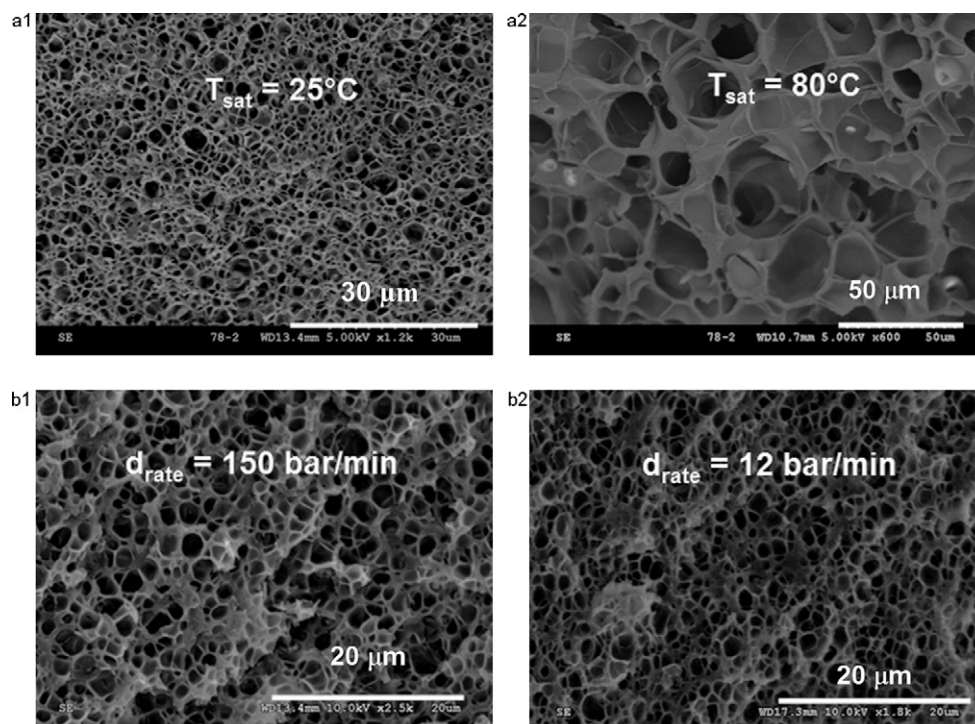


Fig. 3. SEM micrographs showing the morphology evolution in PMMA foams. (a1 and a2) At fixed depressurization rate (60 bar/min). (b1 and b2) At fixed saturation temperature (40 °C).

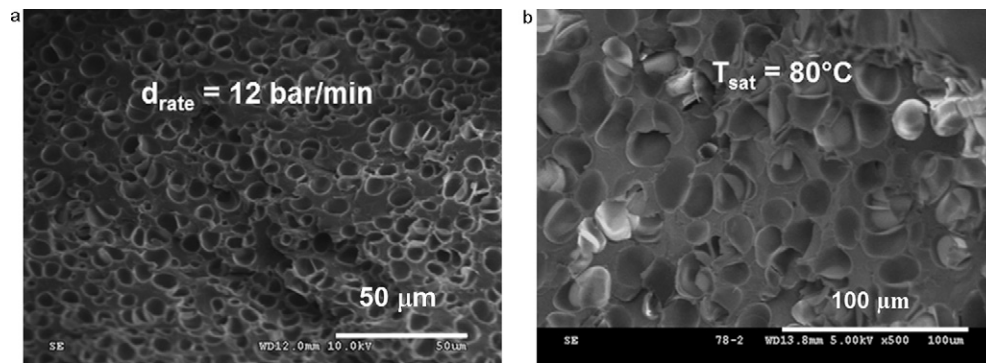


Fig. 4. SEM micrographs showing the morphology evolution in PS foams. (a) At fixed depressurization rate (60 bar/min). (b) At fixed saturation temperature (40 °C).

ering the PMMA + MAM blend, wormlike domains correspond to the separated butylacrylate block of MAM copolymer; whereas in the case of PS + SBM, micelles correspond to the butadiene blocks of SBM terpolymer. Moreover, in the PS + SBM blends, the micelles are large enough to scatter light, and the modified PS is not perfectly transparent (samples were slightly opaque). On the contrary, in the PMMA + MAM blends the MAM copolymer is nanostructured at scales below 100 nm, showing transparent samples. Such nanostructured geometries were reported previously in several works [45,46] in blends from epoxy thermosets with MAM or SBM triblocks.

Figs. 5 and 6 present the cellular morphologies resulting from the main processing parameters for both blends. Again, foaming was not possible in the case of the PS + SBM blend processed at room temperature and 60 bar/min.

Although the SBM copolymer is well dispersed at a sub micro level due to he transparency showed in all the PS + SBM samples, it does not enhances greatly the total CO₂ absorption (see Table 3), while at the same time probably localizing CO₂ in the PMMA blocks of SBM. Indeed these blends are only foaming at temperatures above 50 °C. These results are in accordance with the behaviour of the neat PS showed in Fig. 4, and it can be concluded that the addition of the triblock SBM does not improves the cellular structure at low foaming temperatures.

However, increasing the process temperature up to 80 °C results in a very homogeneous cellular structure (see Fig. 5a), with an average cell size of 30 μm and a high number of nucleated sites. Comparing this micrograph with the neat PS foam obtained in the same conditions (see Fig. 4), it can be seen that at this temperature the addition of the triblock SBM increases the number of nucleated sites with a small variation of the average cell size (from 20 μm in the neat PS foam to 30 μm in the PS + SBM foam).

Also, considering the second foaming route (fixed saturation temperature at 40 °C), the results are presented in previous Fig. 6. As expected, in the PS + SBM blend only foamed structures are reached at low depressurization rates ($d_{rate} = 12 \text{ bar/min}$, see Fig. 6a2). Compared to the neat PS foam (see Fig. 4), the addition of the triblock SBM results in a increase of the cell density, keeping a rather similar value for the average cell size (PS presents an average cell size of 15 μm whereas blend PS + SBM shows an average cell size of 20 μm). So, the effect of the triblock copolymer SBM in the PS matrix is directly focused in an increase in the number of nucleation sites, thus leading directly to higher cell densities, together with a decrease of the coalescence effect.

Secondly considering the PMMA + MAM blends, a completely distinct foaming behaviour is revealed: the addition of the triblock copolymer MAM (10 wt%) into the PMMA matrix results in an improvement of the overall foaming process. In all the temperatures analysed, a microcellular structure is obtained, with an average cell size from 0.3 μm to 5 μm (see Fig. 5b1 and b2), even producing ultra-micro cellular materials when combining low saturation temperatures and high depressurization rates. The fundamental reason is a good compromise between a nano-phase separated and highly CO₂ philic additive with a rubbery block.

Indeed, in the PMMA + MAM blends, the most appreciated effect is seen at high depressurization rates ($d_{rate} = 150 \text{ bar/min}$, see Fig. 6b1), where an ultra-microcellular material is obtained, with an average cell size of 0.3 μm. If depressurization rate is reduced, this effect no longer appears. For example, in the case of the lowest depressurization rate ($d_{rate} = 12 \text{ bar/min}$, see Fig. 6b2), an inhomogeneous cellular structure, with an average cell size ranging from 10 μm to 50 μm, is produced. In this latter case, the nucleation is ruled out by the long growth phase ($\Delta t = 21 \text{ min}$) and also by a secondary nucleation. The PMMA + MAM blends are so far the

Table 4
Main values of cell size ϕ , foam density ρ_f and cell density N_c for the PMMA and PS microcellular foams produced.

	Cell size (μm)	Foam density (g/cm ³)	Cell density (cells/cm ³)	Cell size (μm)	Foam density (g/cm ³)	Cell density (cells/cm ³)
	PMMA			PS		
T_{sat} (°C)						
25	3	1.13	2.41×10^{12}	No foaming occurred	No foaming occurred	No foaming occurred
40	5	1.05	1.56×10^{12}	No foaming occurred	No foaming occurred	No foaming occurred
50	7	0.97	9.42×10^{11}	No foaming occurred	No foaming occurred	No foaming occurred
60	15	0.88	1.41×10^{11}	6	0.94	1.00×10^{12}
70	20	0.83	6.93×10^{10}	12	0.84	2.29×10^{11}
80	25	0.81	3.76×10^{12}	15	0.70	1.92×10^{11}
d_{rate} (bar/min)						
150	2	1.13	8.16×10^{12}	No foaming occurred	No foaming occurred	No foaming occurred
75	4	1.06	2.80×10^{12}	No foaming occurred	No foaming occurred	No foaming occurred
50	5	0.98	2.48×10^{12}	2	1.02	9.00×10^{12}
20	7	0.88	1.39×10^{12}	4	0.99	1.97×10^{12}
12	8	0.85	1.02×10^{12}	13	0.95	9.02×10^{10}

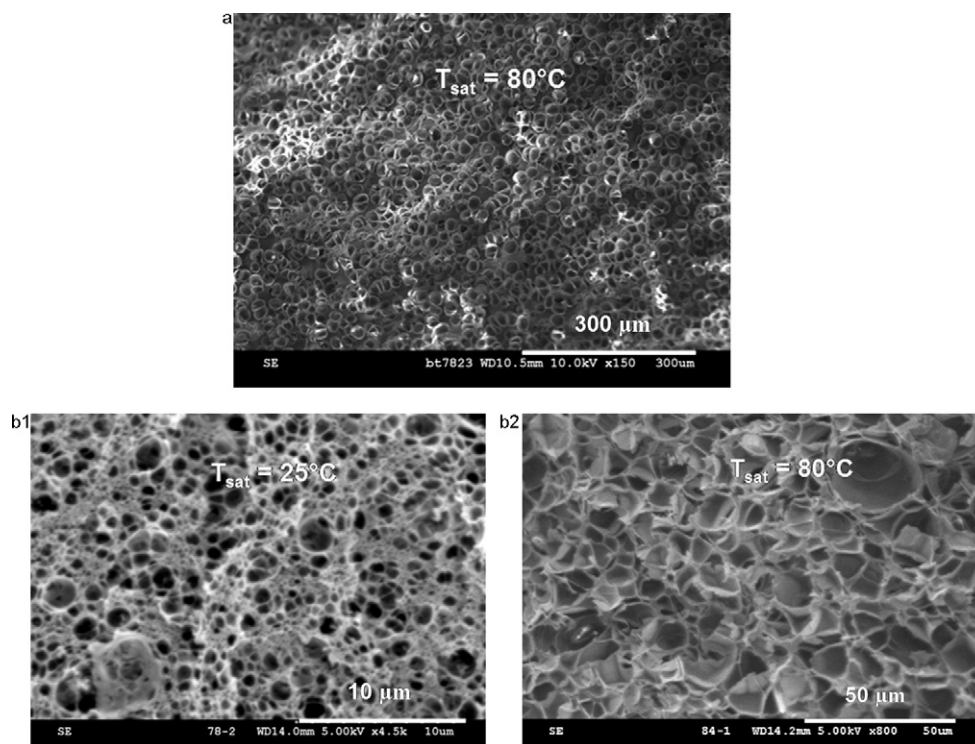


Fig. 5. SEM micrographs of blend foams at fixed depressurization rate (60 bar/min). (a) PS + SBM. (b) PMMA + MAM.

most interesting from of view of ultra micro cellular cells. In Fig. 7 two micrographs of these ultra micro cellular foams produced at 25 °C of saturation temperature and high depressurization rate ($d_{\text{rate}} = 150$ bar/min) are presented.

The overall effect on the main structural parameters for the PS + SBM and PMMA + MAM foams is presented in Table 5.

As a summary, whenever a separated CO_2 -philic is incorporated, i.e. an additive that has more affinity for CO_2 than the matrix, cellular structures nucleate more easily than in the corresponding matrix. The major difference arises from the extent of cellular growth linked to the level of structuration of this additive.

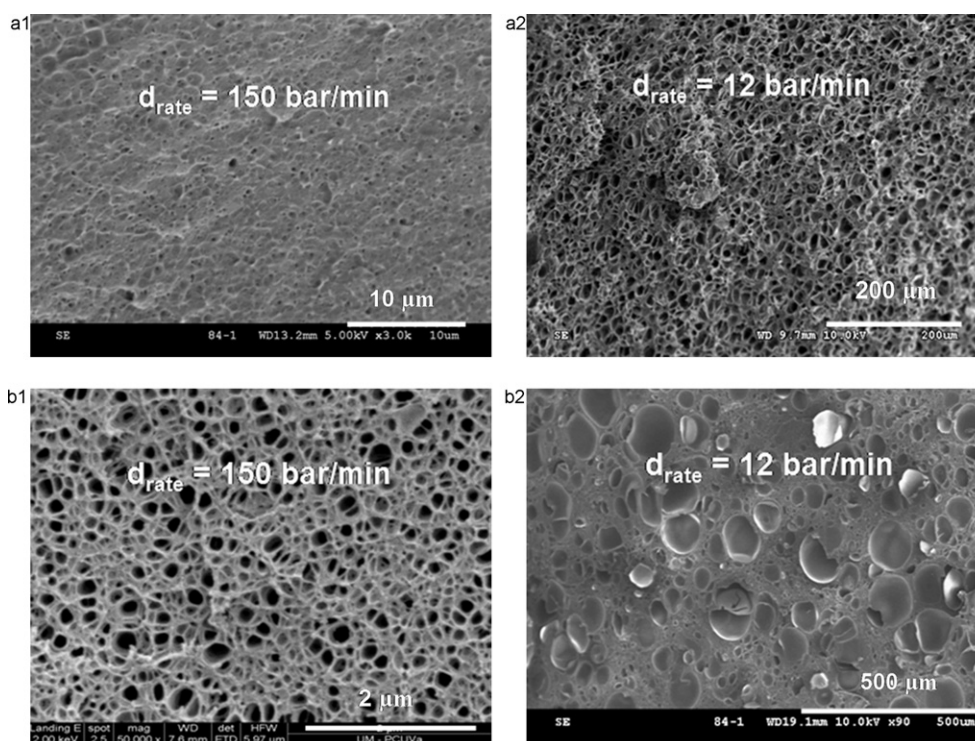


Fig. 6. SEM Micrographs of blend foams at fixed saturation temperature (40 °C). (a) PS + SBM. (b) PMMA + MAM.

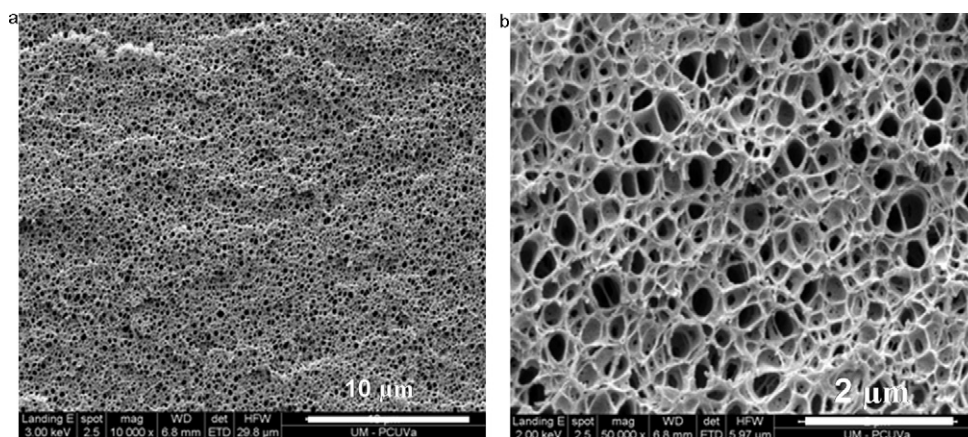


Fig. 7. Example of a PMMA + MAM ultra microcellular foam (saturation temperature room temperature, $d_{\text{rate}} = 150$ bar/min).

Table 5

Main values of cell size ϕ , foam density ρ_f and cell density N_c for the PMMA + MAM and PS + SBM microcellular foams produced.

	PMMA + MAM			PS + SBM		
	Cell size (μm)	Foam density (g/cm^3)	Cell density (cells/cm^3)	Cell size (μm)	Foam density (g/cm^3)	Cell density (cells/cm^3)
T_{sat} ($^{\circ}\text{C}$)						
25	1	1.00	3.74×10^{13}	No foaming occurred	No foaming occurred	No foaming occurred
40	2	0.93	4.81×10^{13}	No foaming occurred	No foaming occurred	No foaming occurred
50	3	0.92	1.53×10^{13}	No foaming occurred	No foaming occurred	No foaming occurred
60	4	0.83	5.31×10^{12}	15	0.94	3.92×10^{10}
70	5	0.78	3.47×10^{12}	20	0.85	3.78×10^{10}
80	7	0.65	1.98×10^{12}	30	0.76	1.75×10^{10}
d_{rate} (bar/min)						
150	0.3	0.71	2.14×10^{14}	No foaming occurred	No foaming occurred	No foaming occurred
75	2	0.65	8.65×10^{13}	No foaming occurred	No foaming occurred	No foaming occurred
50	10	0.63	7.30×10^{11}	No foaming occurred	No foaming occurred	No foaming occurred
20	20	0.60	9.83×10^{10}	12	0.90	1.20×10^{11}
12	80	0.53	1.79×10^9	20	0.82	4.49×10^{10}

A possible scheme of the effect of several nanoadditives in the nucleation and subsequently foaming behaviour is exposed in Fig. 8. Structured block copolymers are one example where the mechanism requires rubbery and CO_2 -philic (nano separated) inner cores acting as CO_2 concentrators and nucleating agents. These cores enable cell growth which may be limited by vitrification of hard poorly CO_2 -philic outer layers. Such objects (Fig. 8) can be introduced in polymer matrices (PS, PMMA, ...) at low levels.

4. Conclusions

Microcellular foaming of amorphous polymers, poly(methyl methacrylate) and poly(styrene) was studied in supercritical

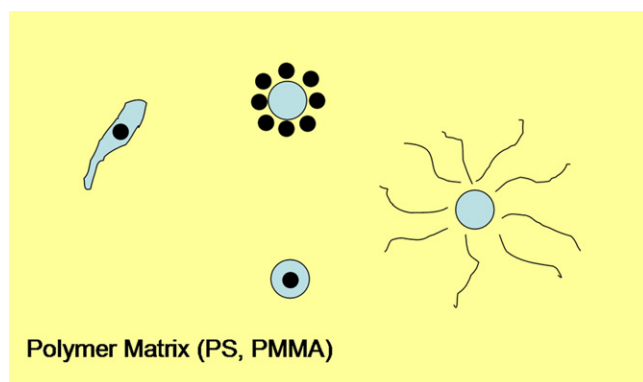


Fig. 8. General scheme for nanodispersed CO_2 -philic additive able to induce micro or nanocellular polymer foaming. The case of nanostructured block copolymers appears e.g. as core/shell particles or as core/hair objects or micelles.

CO_2 via a one-step batch process in the presence of types of additives able to change their foaming behaviour and therefore the porous structures. Triblock polymers (styrene-co-butadiene-co-methylmethacrylate SBM), or (methylmethacrylate-co-butylacrylate-co-methylmethacrylate MAM) were able to form nanostructured blends in PS and PMMA at low contents (10 wt%).

It is shown that block copolymers provide an improvement in reducing the cell size, increasing the cell density mainly due to an increase of the number of rubbery CO_2 -philic nucleation sites, well stabilized in the polymer matrix.

Considering a microcellular foaming route where depressurization rate is fixed (e.g. 60 bar/min) and saturation foaming temperature varies from 25°C to 80°C , the addition of the copolymers did not lead to an increase of the cell density in the polymer blends microcellular foams respect to the neat homopolymer foams. With respect to the neat copolymers analysed, only microcellular structures were obtained using the triblock MAM, whereas SBM did not lead to foams for all the temperatures due to early vitrification and poor CO_2 affinity. The variation of the saturation temperature highlights the role of vitrification of the polymer matrix or the outer layer of the dispersed structured additive.

As a result, block copolymer additives are particularly efficient towards producing micro and nanocellular polymer foams when they act both as nucleating agents, CO_2 concentrators and assist the vitrification of the growing cells (via a rigid poorly CO_2 -philic block). They also offer an improvement in the cell size and cellular homogeneity. For example addition of a triblock MAM copolymer to PMMA resulted in a drastic reduction of the average cell size, producing homogeneous ultra-microcellular foams with an average cell size close to $0.1 \mu\text{m}$ and cell densities of $10^{14} \text{ cells}/\text{cm}^3$ (i.e. close to the nano cellular range).

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